

Week 4 - Team Assignment

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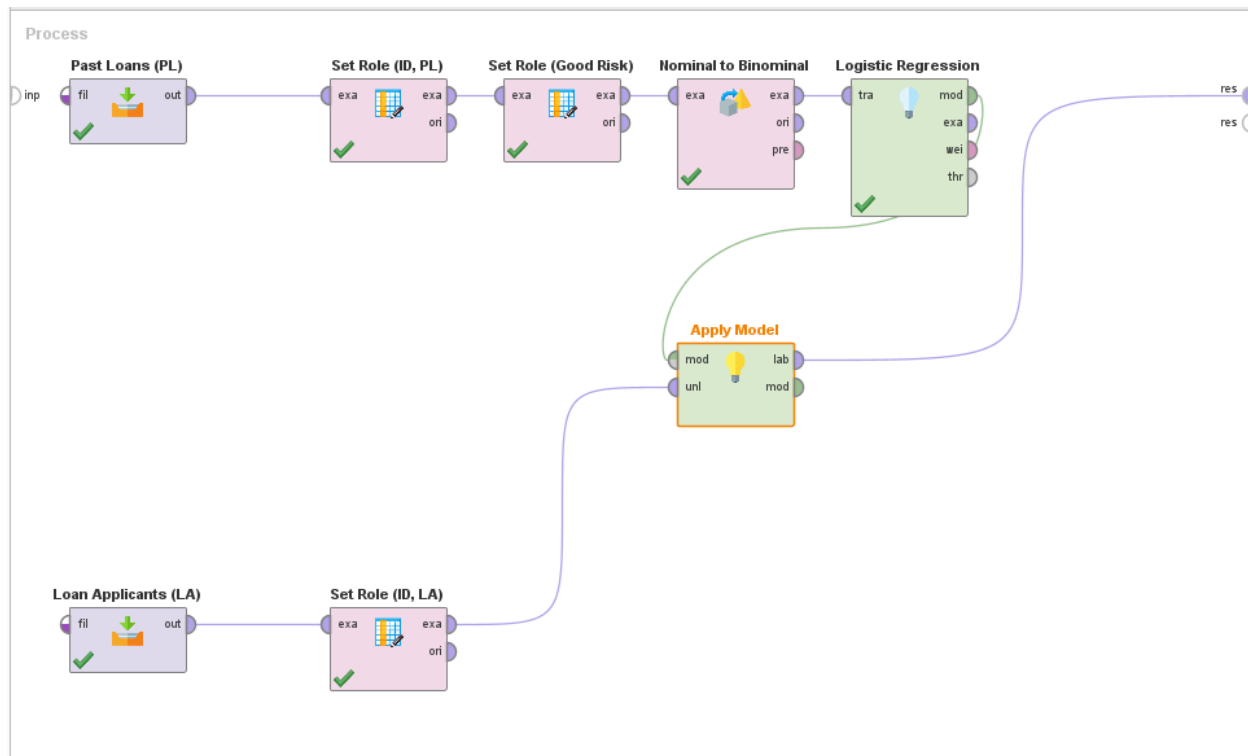
DAT/350: Analyzing And Mining Data

Professor Krishnakumar

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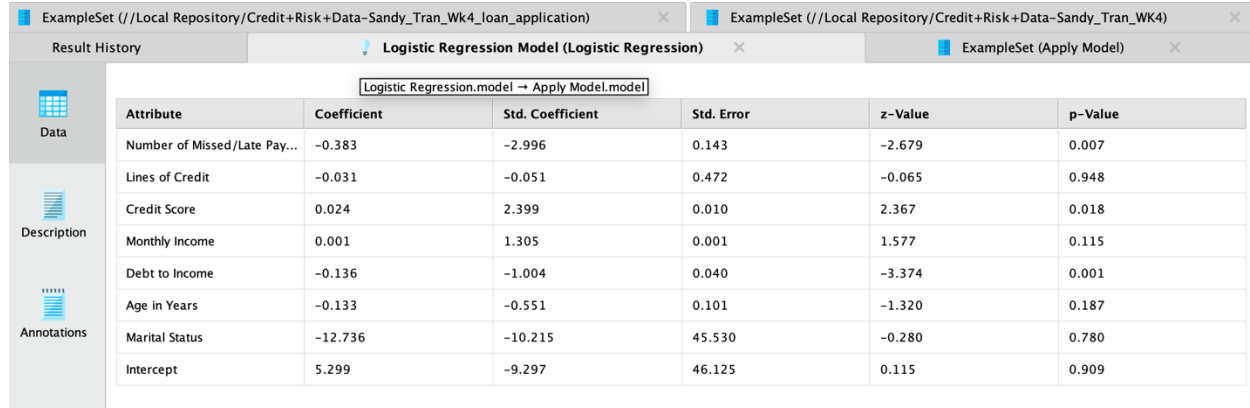
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Image of Logistic Regression Model in Design View:



Explanation: I have put together a model to train whether loan applicants have good or bad risk. The training model (top row) is the "Past Loans" worksheet from the credit risk file from week 2. This model looks at current credit reports, lines of credit, monthly income, age, and number of missed/late payments. This past information is used to train the model to make the determination of whether the applicant is a "good risk" candidate to receive a loan from this institution.

Image of Logistic Regression Model's Properties:



Attribute	Coefficient	Std. Coefficient	Std. Error	z-Value	p-Value
Number of Missed/Late Pay...	-0.383	-2.996	0.143	-2.679	0.007
Lines of Credit	-0.031	-0.051	0.472	-0.065	0.948
Credit Score	0.024	2.399	0.010	2.367	0.018
Monthly Income	0.001	1.305	0.001	1.577	0.115
Debt to Income	-0.136	-1.004	0.040	-3.374	0.001
Age in Years	-0.133	-0.551	0.101	-1.320	0.187
Marital Status	-12.736	-10.215	45.530	-0.280	0.780
Intercept	5.299	-9.297	46.125	0.115	0.909

Explanation: The properties of our logistic regression model consist of our model's measure of variability, standard error values, z-values, and p-values corresponding to each of the attributes. The smaller the p-value of an attribute is, the stronger the predictive power that attribute has, and it is less likely for it to be a random coincidence. According to the properties of our logistic regression model, the strongest predictive independent variables based on the p-values would be the attributes "Debt to Income" with the smallest p-value of 0.001, followed by "Number of Missed/Late Payments" with a p-value of 0.007, and "Credit Score" with a p-value of 0.018. The poorest predictive independent variables based on p-values would be "Lines of Credit" with a p-value of 0.948, followed by "Marital Status" with a p-value of 0.780.

Image of Predictions and Post-Probability Confidence Percentages (Rows 1-25):

Applicant ID	prediction(Good Risk)	confidence(N...	confidence(Yes)	Number of Missed/Late Payments	Lines of Cre...	Credit Score	Monthly Inco...	Debt to Income	Age ...	Marital Status
250162	No	1.000	0.000	13	5	511	3987	36	31	2
337157	No	0.999	0.001	22	4	495	2928	25	34	1
696961	Yes	0.199	0.801	7	5	641	4509	39	38	1
102576	Yes	0.002	0.998	6	6	748	4830	28	33	1
399338	No	0.996	0.004	6	7	799	4598	22	44	2
916894	No	1.000	0.000	28	3	519	4329	37	27	3
332229	Yes	0.133	0.867	9	7	693	4841	43	38	1
591594	No	1.000	0.000	22	3	515	3119	45	39	1
988822	Yes	0.000	1.000	5	6	811	5398	30	38	1
990531	No	0.989	0.011	0	6	709	5637	27	38	2
302120	No	1.000	0.000	18	3	491	3003	28	38	2
851836	Yes	0.000	1.000	3	8	789	6030	34	39	1
465514	Yes	0.000	1.000	0	6	772	5771	24	34	1
203291	Yes	0.001	0.999	8	7	810	5179	33	39	1
183488	No	1.000	0.000	25	4	491	2733	24	39	3
528534	No	1	0	28	5	499	3292	45	28	3
260650	Yes	0.000	1.000	0	8	709	6193	24	37	1
963949	Yes	0.588	0.412	8	5	641	4057	44	39	1
455615	No	1.000	0.000	10	3	541	3932	20	35	2
432768	No	0.821	0.179	14	7	560	3837	20	30	1
673501	No	1.000	0.000	25	5	497	2964	21	34	1
334354	Yes	0.000	1.000	1	8	711	5584	30	43	1
450082	No	0.999	0.001	22	3	515	3408	33	30	1
799506	No	0.961	0.039	16	4	548	3467	24	44	1
839577	Yes	0.002	0.998	4	8	795	5525	49	36	1

Explanation: Most predictions showed extremely high confidence values, especially for applicants predicted as bad risk. For example, many applicants had confidence (no) values above 0.99, suggesting the model saw a strong similarity between their financial profiles and the bad risk patterns in the training data. This is consistent with the training dataset, which likely contained stronger or more consistent signals for identifying bad risk borrowers.

Image of Apply Model's Statistics with the Count of Good vs. Bad Applicants:

✓ Applicant ID	Integer	0	Min 102576	Max 990531	Average 551170.244
✓ prediction(Good Risk)	Binominal	0	Negative No	Positive Yes	Values No (58), Yes (24)
✓ confidence(No)	Real	0	Min 0.000	Max 1	Average 0.730
✓ confidence(Yes)	Real	0	Min 0	Max 1.000	Average 0.270
✓ Number of Missed/Late Payme...	Integer	0	Min 0	Max 29	Average 10.732

The Logistic Regression model generated Good Risk predictions for all 82 loan applicants in the scoring dataset. Out of the 82 applicants, 24 were predicted to be Good Risk (Yes) and 58 were predicted to be Bad Risk (No). The model also produced post-probability confidence percentages for each prediction, which indicate how strongly the model believes an applicant belongs to the predicted category.

Image of Predictions and Post-Probability Confidence Percentages (Rows 31, 36, 58, & 81):

Row No.	Applicant ID	prediction(Good Risk)	confidence(No)	confidence(Yes) ↑	Number of Missed/La...	Lines of Credit	Credit Score	Monthly Inco...	Debt to Inco...	Age in Years
70	834030	No	0.973	0.027	3	8	803	5309	24	43
24	799506	No	0.961	0.039	16	4	548	3467	24	44
62	280600	No	0.920	0.080	3	7	832	5554	24	42
36	373615	No	0.895	0.105	1	6	833	5765	30	33
31	587635	No	0.875	0.125	4	7	836	6018	26	39
20	432768	No	0.821	0.179	14	7	560	3837	20	30
64	247490	No	0.789	0.211	12	4	600	3782	31	35
41	958620	Yes	0.605	0.395	11	6	608	3944	30	33
18	963949	Yes	0.588	0.412	8	5	641	4057	44	39
81	900707	Yes	0.364	0.636	12	7	621	5008	37	32
58	568492	Yes	0.320	0.680	6	6	565	4353	29	41
3	696961	Yes	0.199	0.801	7	5	641	4509	39	38
49	492918	Yes	0.182	0.818	6	5	626	4639	39	35
39	164313	Yes	0.155	0.845	12	4	646	4296	24	35
7	332229	Yes	0.133	0.867	9	7	693	4841	43	38
27	174765	Yes	0.084	0.916	5	6	629	4947	38	40

However, several applicants showed borderline or uncertain predictions, where the confidence percentages were much closer together. For example, Applicant 963949 (Row 18) was predicted as Good Risk with a confidence of 0.412 for Yes and 0.588 for No, indicating the model was not strongly certain about the classification. Similar uncertainty appeared in

Applicants 587635 (Row 31), 373615 (Row 36), 568492 (Row 58), and 900707 (Row 81). These cases suggest that the applicants' financial profiles do not clearly match either the good risk or bad risk patterns in the training data. In a real lending environment, these borderline applicants would likely require additional manual review before making a final decision.